MAX Z

LP · DA · MCDA

COURSE SLIDES · WSQ

Decision-Making and Resource Optimization with Linear Programming

WSQ Course Code: TGS-2023036656 · TSC: Resource Management (WST-SPI-4006-1.1-1)

Conducted by Tertiary Infotech Academy Pte Ltd · UEN 201200696W

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COURSE ADMINISTRATION

Welcome & Housekeeping

Digital Attendance (Mandatory)

1

Three times a day

Take the AM, PM and Assessment digital attendance — mandatory for every WSQ-funded course.

2

Trainer shows the QR

The trainer or administrator displays the SSG digital-attendance QR code (TRAQOM).

3

Scan and submit

Scan the QR with your phone camera and submit your attendance.

4

75% minimum

A minimum of 75% attendance is required to be eligible for assessment and funding.

About the Trainer

?

Your Trainer

General Trainer template —
to be completed by the trainer

NAME

TITLE / DESIGNATION

QUALIFICATIONS

AREAS OF EXPERTISE

TRAINING & INDUSTRY EXPERIENCE

CONTACT

YOUR TRAINER

About the Trainer

AA

Dr Alfred Ang

Principal Trainer
Tertiary Infotech Academy Pte Ltd

ROLE
Principal Trainer, Tertiary Infotech Academy Pte Ltd

EXPERTISE
Operations research, optimisation, data analytics and applied decision science.

DELIVERS
WSQ courses on Linear Programming, decision analysis, statistics and data analytics with Excel.

APPROACH
Hands-on, spreadsheet-led — every method is built and solved live in Microsoft Excel.

ICE-BREAKER

Let's Know Each Other

- Your name, organisation and role.
- How you currently plan or allocate resources (budget, staff, materials, time).
- A decision you make often that you would like to make more objectively.
- Your comfort level with Microsoft Excel (we start from the basics of each tool).

HOUSEKEEPING

Ground Rules

1. Set your mobile phone to silent mode.
2. Participate actively — no question is too small.
3. Mutual respect: agree to disagree.
4. One conversation at a time.
5. Be punctual; return from breaks on time.
6. 75% attendance is required.

Download Your Course Material

lms-tms.tertiaryinfotech.com

Log in → open this course → Course Materials

1. **Trainer Slides (PPT / PDF)**
   The full deck for reference during the open-book assessment.
2. **Learner Guide (PDF)**
   Step-by-step Excel Solver, Decision-Analysis and AHP procedures.
3. **Lesson Plan (PDF)**
   The 2-day schedule with topics, labs and timings.
4. **Lab Files (Excel + CSV)**
   Every lab workbook and sample data set, ready to open in Excel.

Skills Framework Alignment

1

TSC Title

Resource Management

2

TSC Code

WST-SPI-4006-1.1-1

3

Proficiency & duration

WSQ · 2 days (16 hours) · classroom + hands-on Excel practice.

4

Assessed by

Written Assessment (SAQ) for knowledge + Practical Performance for ability.

WHAT YOU WILL KNOW · TESTED BY THE WRITTEN ASSESSMENT

Underpinning Knowledge (K1-K3)

1. **K1 — Resource management, allocation and optimisation tools and techniques.**

2. **K2 — The process to determine resource requirements for the organisation and the business unit.**

3. **K3 — Methods to optimise resource usage.**

WHAT YOU WILL BE ABLE TO DO · TESTED BY THE PRACTICAL PERFORMANCE

Abilities (A1-A9)

1

A1

Develop resource management and allocation strategies.

2

A2

Develop resource allocation plans to support the achievement of strategies.

3

A3

Determine resource needs to support resource allocation planning.

4

A4

Develop resource allocation plans to support business unit activities.

5

A5

Develop procedures and systems to monitor and evaluate resource usage.

6

A6

Review resource usage to determine sufficiency and optimal utilisation.

7

A7

Propose improvements to optimise resource usage.

8

A8

Review resource-management outcomes to ascertain refinements to the resource-allocation plan.

9

A9

Develop resource-management Key Performance Indicators (KPIs).

WHAT YOU'LL ACHIEVE

Learning Outcomes

1

LO1 · Linear Programming

Develop a resource-management strategy and allocation plan using LP and the Excel Solver.

2

LO2 · Decision Analysis

Develop an allocation plan and track resource usage using Expected Value and Expected Utility.

3

LO3 · Multi-Criteria Decision Analysis

Evaluate performance and set KPIs using the scoring board and the Analytic Hierarchy Process.

THREE TOPICS · EIGHT HANDS-ON LABS

Course Outline

1

Topic 1 — Optimisation & Linear Programming

LP models · Excel Solver · binding constraints, shadow prices · sensitivity & scenario analysis. Labs 1-3.

2

Topic 2 — Decision Analysis

Decision trees · Expected Value, EVwPI, EVPI · risk & Expected Utility. Labs 4-5.

3

Topic 3 — Multi-Criteria Decision Analysis

Weighted scoring board · AHP pairwise comparison, consistency · integrated evaluation & KPIs. Labs 6-8.

ORIENTATION

How This Course Works

1. **Hands-on in Excel**
   Every method is built and solved live in Microsoft Excel — bring a laptop.

2. **Eight progressive labs**
   LP, Solver, sensitivity, EV, utility, scoring, AHP and integrated evaluation.

3. **One running example**
   Lucy's Cupcakes, KAZO and supplier / plan cases carry through the course.

4. **Guide + workbooks**
   The Learner Guide has click-by-click steps; each lab ships a ready workbook.

SCHEDULE

Lesson Plan — 2 Days, 8 hours/day

Day 1

- Day 1 — Linear Programming, the Excel Solver and Decision Analysis
  - Admin · Introductions · Skills framework & outcomes
  - Topic 1: Linear Programming & Excel Solver (Labs 1-3)
  - Topic 2 begins: Decision Analysis & EV (Lab 4)
  - 9:00am–6:00pm · 1-hour lunch · tea breaks within

Day 2

- Day 2 — Risk, Multi-Criteria Decision Analysis, AHP and Assessment
  - Topic 2: Risk & Expected Utility (Lab 5)
  - Topic 3: MCDA, AHP & integrated evaluation (Labs 6-8)
  - Briefing for Assessment
  - Final Assessment: WA (SAQ) 1h + PP 1h · TRAQOM

BEFORE THE ASSESSMENT

Briefing for Assessment

1

Two instruments

Written Assessment (SAQ) tests knowledge K1-K3; Practical Performance tests abilities A1-A9.

2

Open book

You may use the slides, Learner Guide and your lab workbooks.

3

Bring a laptop

The Practical Performance is completed in Microsoft Excel; spare laptops are available.

4

Eligibility

75% attendance and a Competent result are required for funding.

FINAL ASSESSMENT

Assessment

1 **Written Assessment (SAQ)**

1 hour · open book · short-answer knowledge questions (K1-K3).

2 **Practical Performance (PP)**

1 hour · open book · hands-on Excel tasks across LP, DA and MCDA (A1-A9).

3 **Result**

Competent / Not-Yet-Competent, with an appeal process available.

4 **Digital attendance**

Take the Assessment digital attendance (TRAQOM) before you start.

01

TOPIC 01

Optimisation and Linear Programming for Resource Management

LP models · decision variables, objective and constraints · Excel Solver · sensitivity and scenario analysis

Key Concepts — Optimisation and Linear Programming for Resource Management

1

Optimisation frames resource management

Choose the allocation of scarce resources that maximises profit or minimises cost, subject to the limits you actually face.

2

A Linear Programming model has three parts

Decision variables (what you choose), an objective function (what you optimise), and constraints (the resource limits).

3

Excel Solver solves the model

The Simplex LP engine searches the feasible region for the corner point that optimises the objective — no manual algebra.

4

Sensitivity analysis makes the plan robust

Shadow prices value each resource; objective and right-hand-side ranges show how far data can move before the plan changes.

5

Diagnostics keep you honest

Infeasible, unbounded, degenerate and multiple-optima outcomes each have a cause and a fix you can recognise.

WHY OPTIMISE

What Is Optimisation in Resource Management?

1

Scarce resources

Budget, staff-hours, materials and machine time are always limited.

2

Competing uses

Every product or activity consumes resources and returns a benefit.

3

One best allocation

Optimisation finds the allocation that maximises benefit (or minimises cost) within the limits.

4

Linear Programming

When benefit and usage are proportional, the problem is a Linear Program — solvable exactly.

WHERE LP APPLIES

The Resource-Allocation Problem

1

Product mix

How many of each product to make with limited materials and labour?

2

Budget allocation

How to split a fixed budget across projects for the highest return?

3

Staff scheduling

How to cover demand with the fewest staff-hours?

4

Blending / transport

How to mix inputs or route supply at least cost?

THE THREE PARTS

Anatomy of a Linear Programming Model

DECISION VARIABLES	$X1, X2, \ldots \geq 0$	what you choose
OBJECTIVE FUNCTION	Max (or Min) $Z = c1 \cdot X1 + c2 \cdot X2 + \ldots$	what you optimise
CONSTRAINTS	$a1 \cdot X1 + a2 \cdot X2 + \ldots \leq b$	the resource limits

READ IT AS choose the variables that optimise the objective without breaking any constraint.

Linear Programming — Standard Form

OBJECTIVE	**Maximise** $Z = c^T x$	linear in the variables
SUBJECT TO	$A \cdot x \le b$	resource capacity limits
SIGN	$x \ge 0$	non-negativity

READ IT AS every LP is decision variables x, a linear objective c, linear constraints $A \cdot x \le b$, and $x \ge 0$.

The Feasible Region & the Corner-Point Theorem

1. **Feasible region**
   The set of all allocations that satisfy every constraint — a convex polygon.

2. **Corners = vertices**
   Each corner is where constraint lines cross.

3. **Optimum at a corner**
   If an optimum exists, at least one optimal solution sits at a corner point.

4. **Why Solver is fast**
   Simplex only needs to check corners, not the infinite interior.

WHEN LP IS VALID

The Five Assumptions Behind Every LP

1

Proportionality

Doubling a variable doubles its contribution and its resource use.

2

Additivity

Total profit / usage is the sum across variables — no interaction terms.

3

Divisibility

Fractional values are allowed (37.5 batches); use integer LP if not.

4

Certainty

Coefficients and limits are known constants (decision analysis handles uncertainty).

5

Non-negativity

You cannot produce a negative quantity: $X \geq 0$.

THE PROBLEM · TWO PRODUCTS, FOUR LIMITS

Worked Scenario — Lucy's Cupcakes

Resource / item	Vanilla X1	Chocolate X2	Available
Profit per unit ($)	2	3	— maximise —
Baking powder (kg)	0.1	0.2	10
Man-hours (hr)	0.05	0.1	10
Vanilla demand (units)	1	0	25
Chocolate demand (units)	0	1	60

Profit to maximise

EVIDENCE Lucy bakes vanilla (X1) and chocolate (X2) cupcakes for a fair. How many of each maximises profit within the baking-powder, man-hour and demand limits?

Formulating the Model — Variables & Objective

DECISION VARIABLES

$$X1 = \text{vanilla}, \quad X2 = \text{chocolate}$$

units to bake

OBJECTIVE

$$\text{Max} \quad Z = 2 \cdot X1 + 3 \cdot X2$$

total profit ($)

READ IT AS each vanilla earns $2, each chocolate $3 — maximise the weighted sum.

STEP 3 · EACH RESOURCE BECOMES A LINE

Formulating the Model — The Four Constraints

BAKING POWDER	$0.1 \cdot X1 + 0.2 \cdot X2 \leq 10$	kg available
MAN-HOURS	$0.05 \cdot X1 + 0.1 \cdot X2 \leq 10$	hours available
DEMAND CAPS	$X1 \leq 25, \quad X2 \leq 60$	units sellable
NON-NEGATIVITY	$X1, X2 \geq 0$	no negative output

READY TO SOLVE

The Complete LP Model

MAXIMISE	$Z = 2 \cdot X1 + 3 \cdot X2$	profit
SUBJECT TO	$0.1 \cdot X1 + 0.2 \cdot X2 \le 10$ and $0.05 \cdot X1 + 0.1 \cdot X2 \le 10$	resources
AND	$X1 \le 25$, $X2 \le 60$, $X1, X2 \ge 0$	demand & sign

READ IT AS this is exactly what you enter on the worksheet — coefficients, SUMPRODUCT and limits.

Turning Words into Constraints

1. **'At most' / 'no more than'**
   A '<=' constraint — a resource ceiling.

2. **'At least' / 'minimum'**
   A '>=' constraint — a requirement to meet.

3. **'Exactly' / 'must equal'**
   An '=' constraint — a fixed target.

4. **'Cannot be negative'**
   X >= 0 — non-negativity.

Resource Efficiency — Read Before You Solve

Profit per kg of the binding resource (baking powder) tells you which product to favour first.

```
# profit per kg baking powder
Vanilla:   $2 / 0.1kg = $20/kg
Chocolate: $3 / 0.2kg = $15/kg

# vanilla is more powder-
# efficient, so max it first
X1 = 25   (demand cap)
# then fill powder with X2
X2 = (10-2.5)/0.2 = 37.5
```

Efficiency ratio

Profit divided by resource used per unit.

Vanilla wins on powder

$20/kg beats $15/kg — favour vanilla.

But demand caps it

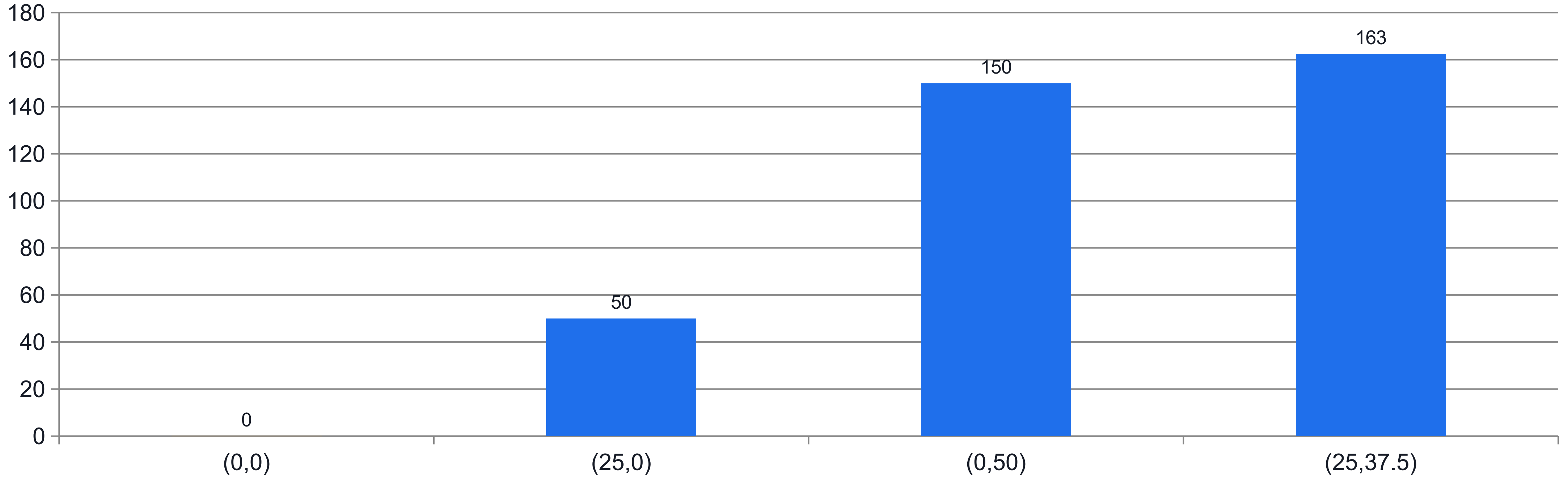
Only 25 vanilla sell; chocolate fills the rest.

Matches Solver

The intuition gives Solver's exact answer.

TOPIC 01 · THE OPTIMUM IS A CORNER

Every Corner of the Feasible Region



WHAT THE DATA SHOWS

Profit is evaluated only at the corners of the feasible region. The corner (25, 37.5) gives the maximum, $162.50 — the corner-point theorem in action.

Iso-Profit Lines & the Optimal Corner

1

Iso-profit line

All (X1,X2) giving equal profit form a line: $2X_1+3X_2$ = constant.

2

Slide it outward

Higher profit moves the line away from the origin.

3

Last touch wins

The optimum is the last corner the line touches inside the feasible region.

4

Always a vertex

That last-touch point is a corner — never the interior.

Formulate a Linear Programming Model

LAB 1

A1, A3, K1 — develop an allocation strategy by translating a resource problem into an LP model

Turn Lucy's Cupcakes resource problem into a Linear Programming model: name the decision variables, write the profit objective, and express every resource limit as a linear constraint.

YOU'LL BUILD

A complete LP model (objective + four constraints) laid out on a formula-driven Excel worksheet, ready to solve.

TOOLCHAIN

Microsoft Excel, lab CSV data

DONE WHEN

The worksheet shows a single objective cell (E6) and four constraint LHS cells (E9:E12), each a SUMPRODUCT of coefficients and the changing cells. With X1=X2=0 the objective reads 0 and every constraint LHS reads 0.

LAB 1 · HOW IT RUNS (NO CLICKS — SEE THE LEARNER GUIDE)

Formulate a Linear Programming Model

1
Read the problem
List products, profit per…

→

2
Name variables
Define X1, X2 — the units of…

→

3
Write the objective
Max Z = 2·X1 + 3·X2 (total…

→

4
Write constraints
One linear inequality per…

→

5
Lay out the sheet
Coefficients, SUMPRODUCT for…

THE POINT

A1, A3, K1 — develop an allocation strategy by translating a resource problem into an LP model

Lab 1 Evidence — The Model on the Worksheet

Row / item	Vanilla X1	Chocolate X2	Used (LHS)		Limit
Units to bake	0	0	← changing cells		
Profit / unit ($)	2	3	Z =SUMPRODUCT	→	MAX
Baking powder (kg)	0.1	0.2	=SUMPRODUCT	≤	10
Man-hours (hr)	0.05	0.1	=SUMPRODUCT	≤	10
Vanilla demand	1	0	=SUMPRODUCT	≤	25
Chocolate demand	0	1	=SUMPRODUCT	≤	60

Changing cells X1,X2   Objective Z   Constraint LHS =SUMPRODUCT

EVIDENCE The objective and every constraint are SUMPRODUCT formulas of the changing cells — so Solver can vary X1, X2 and recompute instantly.

LAB 1 · VERIFY & DIAGNOSE

Formulate a Linear Programming Model

✅ Expected result

The worksheet shows a single objective cell (E6) and four constraint LHS cells (E9:E12), each a SUMPRODUCT of coefficients and the changing cells. With X1=X2=0 the objective reads 0 and every constraint LHS reads 0.

IF IT DOESN'T WORK — DIAGNOSE

Objective is a fixed number, not a formula

You typed the profit instead of =SUMPRODUCT(...). Solver can only optimise a cell that depends on the changing cells.

Constraint uses absolute vs relative refs wrongly

Lock the changing cells with $C$4:$D$4 so the formula copies down correctly.

A resource limit is on the wrong side

The limit (RHS) is a constant in column G; the used amount (LHS) is the SUMPRODUCT in column E. Do not swap them.

THE ENGINE (CONCEPT — STEPS ARE IN THE LEARNER GUIDE)

What the Excel Solver Does

1

Reads your model

The objective cell, the changing cells and the constraint cells you defined.

2

Searches corners

The Simplex LP engine moves corner-to-corner, improving the objective each step.

3

Stops at the optimum

When no adjacent corner is better, it reports the optimal solution.

4

Explains it

Answer and Sensitivity reports show binding constraints, slack and shadow prices.

MAPPING THE MODEL ONTO SOLVER

The Solver Model — Which Cell Plays Which Role

Solver setting	Cell(s)	Meaning
Set Objective → Max	E6 (Z)	Total profit =SUMPRODUCT(profit, units)
By Changing Cells	C4:D4	The units X1, X2 Solver may vary
Subject to Constraints	E9:E12 ≤ G9:G12	Resource & demand limits
Make Variables ≥ 0	C4:D4 ≥ 0	Non-negativity
Solving Method	Simplex LP	The model is linear

Objective cell   Changing cells

EVIDENCE Solver needs four things: an objective cell to optimise, changing cells to vary, constraint cells to respect, and the Simplex LP engine.

HOW THE OPTIMUM IS FOUND

Simplex moves corner to corner.

From a starting vertex, the Simplex LP engine steps to the adjacent corner that most improves profit — repeating until no better corner exists.

TOPIC 01 · HANDS-ON LAB 2

Solve for Maximum Profit with Excel Solver

LAB 2

A2, K1 — develop an allocation plan that maximises profit using the Excel Solver

Run the Simplex LP engine in Excel Solver on the Lucy's Cupcakes model to find the profit-maximising production plan and produce the Answer Report.

YOU'LL BUILD

The optimal plan X1=25, X2=37.5 giving Z=$162.50, plus a Solver Answer Report showing binding and non-binding constraints.

TOOLCHAIN

Microsoft Excel, Solver add-in

DONE WHEN

The changing cells read X1=25 and X2=37.5, the objective E6 reads 162.5, and the Answer Report lists baking powder and vanilla-demand as Binding (slack 0) while man-hours and chocolate-demand are Not Binding (slack 5 and 22.5).

LAB 2 · HOW IT RUNS (NO CLICKS — SEE THE LEARNER GUIDE)

Solve for Maximum Profit with Excel Solver

1

Set the objective

Point Solver at the profit…

→

2

Set changing cells

The units C4:D4 are what…

→

3

Add constraints

E9:E12 ≤ G9:G12, and…

→

4

Pick the engine

Simplex LP — the model is…

→

5

Solve & report

Keep the solution and…

THE POINT

A2, K1 — develop an allocation plan that maximises profit using the Excel Solver

Lab 2 Evidence — The Solved Worksheet

Row / item	Vanilla X1	Chocolate X2	Used (LHS)		Limit / status
Units to bake (optimal)	25	37.5	← Solver result		
Profit / unit ($)	2	3	Z = 162.50	→	MAX ✓
Baking powder (kg)	0.1	0.2	10.0	≤	10 binding
Man-hours (hr)	0.05	0.1	5.0	≤	10 slack 5
Vanilla demand	1	0	25	≤	25 binding
Chocolate demand	0	1	37.5	≤	60 slack 22.5

Max profit

Binding

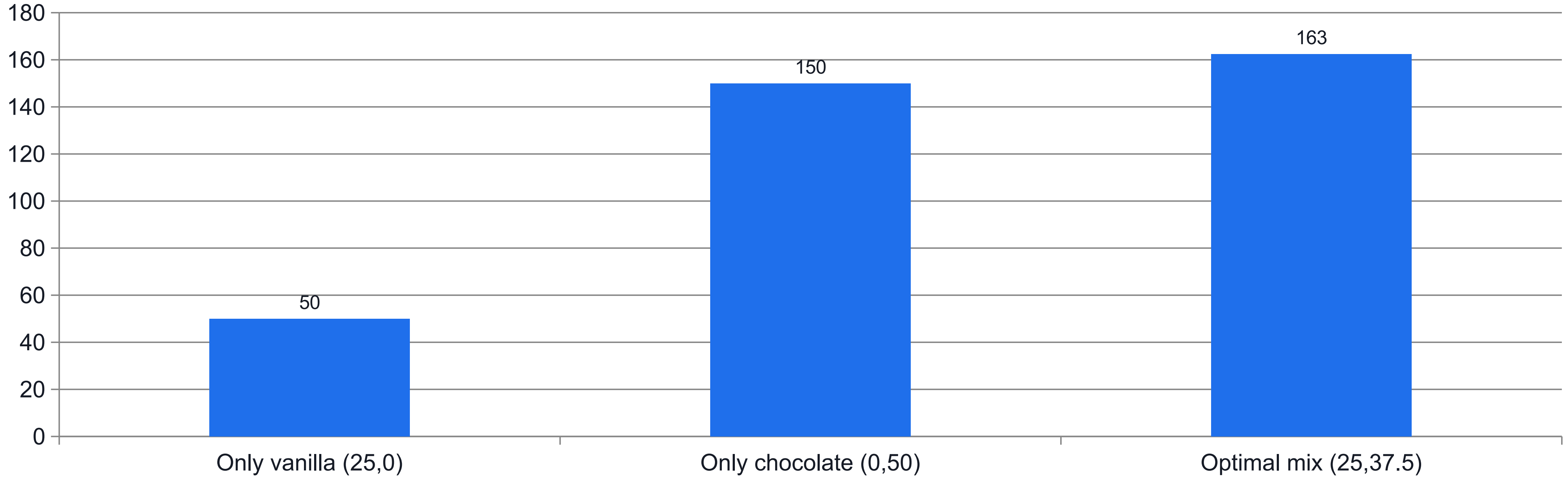
EVIDENCE Baking powder and vanilla demand are fully used (binding); man-hours and chocolate demand have spare capacity (slack).

Reading the Answer Report — Binding vs Non-Binding

Constraint	Final value	Status	Slack
Baking powder (kg)	10.0	Binding	0
Man-hours (hr)	5.0	Not Binding	5
Vanilla demand (units)	25	Binding	0
Chocolate demand (units)	37.5	Not Binding	22.5

WHY IT MATTERS a binding constraint has zero slack and limits profit — it earns a shadow price. Non-binding constraints have spare capacity and add no value.

Lab 2 Evidence — Why the Mix Beats a Single Product



WHAT THE DATA SHOWS

Making only one product wastes a resource. The optimal mix earns $162.50 — more than either pure plan.

LAB 2 · VERIFY & DIAGNOSE

Solve for Maximum Profit with Excel Solver

✅ Expected result

The changing cells read X1=25 and X2=37.5, the objective E6 reads 162.5, and the Answer Report lists baking powder and vanilla-demand as Binding (slack 0) while man-hours and chocolate-demand are Not Binding (slack 5 and 22.5).

IF IT DOESN'T WORK — DIAGNOSE

Solver is not on the Data tab

The add-in is not loaded — repeat the Add-ins step and tick 'Solver Add-in'.

'Solver could not find a feasible solution'

A constraint is impossible or a sign is reversed. Check every LHS ≤ RHS and that the RHS limits are positive.

'The Objective Cell values do not converge' (unbounded)

A binding resource constraint is missing, so profit can grow without limit. Add the omitted constraint.

Reading the Answer Report Line by Line

The Solver Answer Report has three blocks: the objective, the variable cells, and the constraints.

```
OBJECTIVE CELL (Max)
  Z:  0  ->  162.5

VARIABLE CELLS
  X1: 0 -> 25
  X2: 0 -> 37.5

CONSTRAINTS
  Powder 10  Binding  slack 0
  Hours   5  NotBind  slack 5
  DemX1  25  Binding  slack 0
  DemX2 37.5 NotBind  slack 22.5
```

Objective block

Original vs final Z — profit rose from 0 to 162.5.

Variable block

The optimal X1, X2 Solver chose.

Constraint status

Binding = fully used (slack 0); Not Binding = spare.

Where to act

Binding constraints are the bottlenecks worth relaxing.

Interpreting Slack in Business Terms

1 **Slack = spare capacity**
Man-hours slack 5 means 5 hours sit idle at the optimum.

2 **Zero slack = bottleneck**
Baking powder slack 0 means it is fully consumed and caps profit.

3 **Don't buy slack resources**
Extra man-hours add nothing while 5 hr are idle.

4 **Target the bottleneck**
More baking powder is what unlocks more profit.

Excel Solver vs Solving by Hand

Aspect	By hand (graphical / simplex)	Excel Solver
Variables	2-3 before it gets hard	Hundreds
Effort	Manual algebra, error-prone	Automatic
Reports	You derive them	Answer + Sensitivity built in
Best for	Understanding the method	Real problems

WHY IT MATTERS learn the mechanism by hand on small models; use Solver for anything real.

DUAL VALUES

Shadow Prices — The Value of a Resource

SHADOW PRICE	Δ optimal profit per +1 unit of a resource	marginal value
BINDING → POSITIVE	A fully-used resource has a shadow price > 0	worth buying more
SLACK → ZERO	A resource with spare capacity is worth 0 more	don't buy more

READ IT AS baking powder is binding, so one more kg is worth $15 of profit; man-hours have slack, so worth $0.

Sensitivity — How Far Can the Data Move?

1. **Objective ranges**
   How much a profit coefficient can change before the optimal plan (the basis) changes.

2. **RHS ranges**
   How much a resource limit can change while its shadow price stays valid.

3. **Reduced cost**
   How much an unused product's profit must rise before it enters the plan.

4. **Why it matters**
   Prices and supplies move — sensitivity tells you when to re-plan and when not to.

TOPIC 01 · HANDS-ON LAB 3

Sensitivity and Scenario Analysis

LAB 3

A2, A3, A6 — review sufficiency and optimal utilisation using the Sensitivity Report and what-if scenarios

Read the Solver Sensitivity Report to value each resource (shadow prices) and find the ranges over which the plan holds, then run what-if scenarios that change a resource limit or a profit rate.

YOU'LL BUILD

A Sensitivity Report with shadow prices and allowable ranges, plus a small scenario table comparing profit as baking powder and profit rates change.

TOOLCHAIN

Microsoft Excel, Solver add-in, Scenario Manager / Data Table

DONE WHEN

The Sensitivity Report shows baking powder with a positive shadow price and man-hours with a $0 shadow price; increasing baking powder within its allowable increase raises profit by about $15 per kg, and the scenario table shows profit rising as the baking-powder limit increases.

LAB 3 · HOW IT RUNS (NO CLICKS — SEE THE LEARNER GUIDE)

Sensitivity and Scenario Analysis

1. **Re-solve with the report**
   Generate the Sensitivity…
2. **Value the resources**
   Shadow price = extra profit…
3. **Read the ranges**
   Allowable increase/decrease…
4. **Run scenarios**
   Change baking powder or a…
5. **Recommend**
   Buy more of the resource with…

THE POINT

A2, A3, A6 — review sufficiency and optimal utilisation using the Sensitivity Report and what-if scenarios

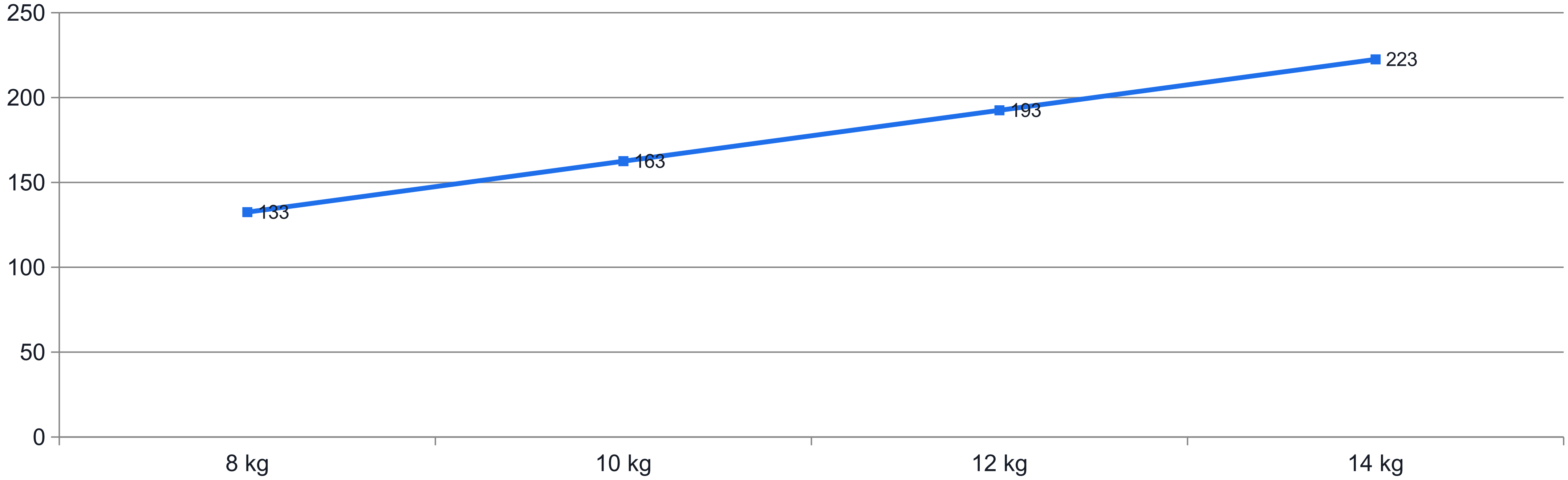
Lab 3 Evidence — The Sensitivity Report

Resource / limit	Shadow price	Binding?	Interpretation
Baking powder (kg)	$15.00	Yes	+1 kg → +$15 profit (within range)
Man-hours (hr)	$0.00	No	5 hr spare — more adds nothing
Vanilla demand	$0.50	Yes	+1 unit cap → +$0.50 profit
Chocolate demand	$0.00	No	22.5 spare — not the bottleneck

WHY IT MATTERS spend on the binding resource with the highest shadow price (baking powder), but only within its allowable increase — beyond that, re-solve.

LAB 3 · WHAT-IF ANALYSIS

Lab 3 Evidence — Scenario: Profit vs Baking Powder



WHAT THE DATA SHOWS

Each extra kg adds exactly $15 — the shadow price — up to ~14.5 kg, where chocolate demand becomes the new bottleneck and the slope changes.

Sensitivity and Scenario Analysis

✅ Expected result

The Sensitivity Report shows baking powder with a positive shadow price and man-hours with a $0 shadow price; increasing baking powder within its allowable increase raises profit by about $15 per kg, and the scenario table shows profit rising as the baking-powder limit increases.

IF IT DOESN'T WORK — DIAGNOSE

No Sensitivity Report is produced

You must select Simplex LP; the GRG Nonlinear engine gives a different report. Re-solve with Simplex LP.

All shadow prices are zero

No constraint is binding — the solution is interior or the model is under-constrained. Re-check the resource limits.

Shadow price used outside its range

A shadow price is only valid within the allowable increase/decrease. Beyond it, re-solve rather than extrapolate.

Shadow Price in Action — Should Lucy Buy More Powder?

The shadow price of baking powder is $15/kg. Compare it to the price of buying more.

```
# baking powder shadow price
= $15 per extra kg

# supplier sells powder at
= $6 per kg

# net gain per extra kg
= 15 - 6 = $9   (buy!)

# valid to about +4.5 kg,
# then demand binds -> re-solve
```

Shadow price

Each extra kg is worth $15 of profit while binding.

Compare to cost

At $6/kg, buying adds $9/kg net profit.

Respect the range

Valid only within the allowable increase (~4.5 kg).

Then re-solve

Beyond the range a different constraint binds.

Diagnosing a Solver Result

Did Solver return a finite optimum?

Yes — read the reports

Record the plan, then read the Answer & Sensitivity reports for binding constraints and shadow prices.

No — diagnose the model

Infeasible, unbounded or degenerate — each has a distinct cause and fix (next slide).

a healthy LP returns one finite optimum. Anything else is a modelling signal, not a dead end.

LP Failure Diagnostics

⚠ Infeasible — no solution

Fix: Constraints contradict (e.g. demand > capacity). Relax or correct the conflicting limit.

⚠ Unbounded — profit → ∞

Fix: A binding resource constraint is missing. Add the omitted capacity limit.

⚠ Multiple optima

Fix: A non-basic reduced cost is zero — a whole edge is optimal. Any of the tied plans is valid.

⚠ Degenerate / cycling

Fix: A constraint is redundant (a tie in the ratio test). Remove duplicates or use Simplex anti-cycling.

Divisibility — Continuous vs Integer LP

1

Continuous LP

Fractions allowed: 37.5 batches is fine when output is divisible.

2

Integer LP

Whole units only — people, vehicles and machines cannot be fractional.

3

Add integer constraints

In Solver, add X = integer; it switches to branch-and-bound.

4

Cost of integers

Integer models solve slower and can lose a little profit vs the LP relaxation.

Common LP Formulation Mistakes

⚠ Objective typed as a number

Fix: Make it =SUMPRODUCT so it depends on the changing cells.

⚠ Constraint direction reversed

Fix: Resource and demand caps are LHS <= RHS, not >=.

⚠ Mixed units

Fix: Keep kg with kg and hours with hours; convert before you model.

⚠ Forgot non-negativity

Fix: Tick 'Make Unconstrained Variables Non-Negative' or add X >= 0.

Recap — Optimisation & Linear Programming

- An LP model = decision variables + a linear objective + linear constraints, with $X \geq 0$.
- Excel Solver (Simplex LP) finds the profit-maximising corner: Lucy's optimum is $X1=25$, $X2=37.5$, $Z=\$162.50$.
- Binding constraints (baking powder, vanilla demand) limit profit and earn shadow prices; slack means spare capacity.
- Sensitivity and scenario analysis show how far data can move and which resource to buy first.
- Infeasible, unbounded, multiple-optima and degenerate results each have a recognisable cause and fix.

02

TOPIC 02

Decision Analysis for Resource Usage

Decision trees · Expected Value (EV, EVwPI, EVPI) · risk and Expected Utility

Key Concepts — Decision Analysis for Resource Usage

1

Decisions under uncertainty need structure

A decision problem is a set of decisions, uncertain states of nature with probabilities, and a payoff for every combination.

2

Decision trees make the structure visible

Square decision nodes branch into your choices; round chance nodes branch into states — you fold back to the best expected payoff.

3

Expected Value ranks the alternatives

EV = Σ probability × payoff. The decision with the highest expected payoff is the EV-optimal choice.

4

Perfect information has a price

EVPI = EVwPI − EV is the most you should ever pay to remove the uncertainty before deciding.

5

Expected Utility captures risk attitude

When outcomes are large or risky, replacing money with utility separates a risk-averse choice from a risk-neutral one.

Decision-Making Under Uncertainty

1 **Decisions**
The options you control — e.g. which expansion to build.

2 **States of nature**
The uncertain future — e.g. a favourable, neutral or unfavourable market.

3 **Probabilities**
Your best estimate of each state's likelihood (they sum to 1).

4 **Payoffs**
The result of every decision-and-state combination.

STRUCTURE FIRST · PAYOFFS IN $'000

Anatomy of a Decision — The Payoff Table

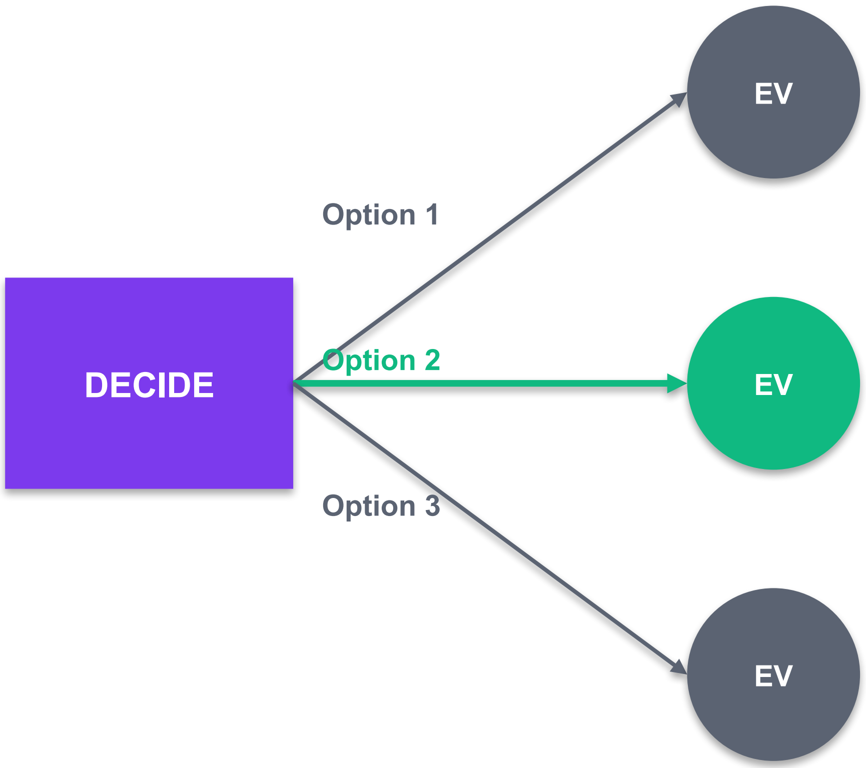
Decision \ State	Favourable	Neutral	Unfavourable
Probability	0.3	0.5	0.2
Option 1 — Large	400	360	100
Option 2 — Medium	350	380	250
Option 3 — Small	300	390	200

State probabilities (sum to 1)

EVIDENCE Rows are the decisions you control; columns are the uncertain states with probabilities. Every cell is a payoff.

SQUARE = DECISION · CIRCLE = CHANCE

The Decision Tree — Nodes & Branches



Fav:0.3→400 Neu:0.5→360 Unf:0.2→100

EV = 320

Fav:0.3→350 Neu:0.5→380 Unf:0.2→250

EV = 345 ← choose

Fav:0.3→300 Neu:0.5→390 Unf:0.2→200

EV = 325

Fold back: at each chance node compute EV = Σ p·payoff; at the decision node pick the branch with the highest EV.

HOW TO CHOOSE

Decision Criteria — With and Without Probabilities

Criterion	Uses probabilities?	Picks	Attitude
Maximax	No	Best of the best payoffs	Optimistic
Maximin	No	Best of the worst payoffs	Pessimistic
Minimax regret	No	Least worst-case regret	Cautious
Expected Value	Yes	Highest Σ p·payoff	Risk-neutral

WHY IT MATTERS with reliable probabilities, Expected Value is the standard criterion — it uses all the information in the payoff table.

THE CORE FORMULA

Expected Value

EXPECTED VALUE

$$EV = \Sigma \; p_i \cdot x_i$$

probability × payoff, summed

DECISION RULE

Choose the option with the highest EV

the EV-optimal choice

READ IT AS EV is the long-run average payoff if the decision were repeated many times.

Valuing Information — EVwPI & EVPI

EV WITH PERFECT INFORMATION

$$\text{EVwPI} = \Sigma\, p_i \cdot (\text{best payoff in state } i)$$

if you knew the future

EV OF PERFECT INFORMATION

$$\text{EVPI} = \text{EVwPI} - \text{best EV}$$

the value of knowing

READ IT AS EVPI is the most you should ever pay for perfect market information before deciding.

Decision Tree vs Payoff Table

1

Payoff table

A compact grid of decisions x states — best for a single decision.

2

Decision tree

Shows the sequence and chance nodes — best for staged decisions.

3

Same maths

Both use EV = Sum(p x payoff) and fold-back.

4

This course

We use the table to compute and the tree to visualise.

TOPIC 02 · SOLVING THE TREE

Folding Back the Decision Tree

Work right to left: compute each option's EV at its chance node, then pick the best at the decision node.

```
# EV = sum(p * payoff)  $'000
Option 1:
  .3*400+.5*360+.2*100 = 320
Option 2:
  .3*350+.5*380+.2*250 = 345
Option 3:
  .3*300+.5*390+.2*200 = 325

# decision node: pick max
max(320,345,325)=345 -> Opt2
```

Chance nodes first

EV = Sum(p x payoff) at each round node.

Then the decision

At the square node choose the highest EV.

Option 2 wins

345 beats 320 and 325.

This is fold-back

The standard decision-tree solution method.

Maximax, Maximin & Expected Value on KAZO

Criterion	Option 1	Option 2	Option 3	Picks
Maximax (best case)	400	380	390	Option 1 (400)
Maximin (worst case)	100	250	200	Option 2 (250)
Expected Value	320	345	325	Option 2 (345)

WHY IT MATTERS Option 1 looks best only to a pure optimist; the risk-aware maximin and the probability-based EV both favour Option 2.

Expected Value, EVwPI and EVPI

LAB 4

A4, A5, K2 — determine resource needs and monitor usage by ranking options on expected payoff

Build a decision table for KAZO's expansion choice, compute the Expected Value of each option, then find the Expected Value with Perfect Information and the Expected Value of Perfect Information.

YOU'LL BUILD

A formula-driven decision table that ranks three options by EV (best EV=$345k), and computes EVwPI=$365k and EVPI=$20k.

TOOLCHAIN

Microsoft Excel

DONE WHEN

The EV column shows Option 1 = 320, Option 2 = 345, Option 3 = 325; the best-EV flag points to Option 2; EVwPI reads 365 and EVPI reads 20 (all in $'000).

LAB 4 · HOW IT RUNS (NO CLICKS — SEE THE LEARNER GUIDE)

Expected Value, EVwPI and EVPI

1. **Structure the decision**
   Three options × three market…

2. **Enter the payoffs**
   A payoff for every…

3. **Compute EV per option**
   EV = Σ probability × payoff…

4. **Pick the best EV**
   The highest EV is the…

5. **Value information**
   EVwPI takes the best payoff…

THE POINT

A4, A5, K2 — determine resource needs and monitor usage by ranking options on expected payoff

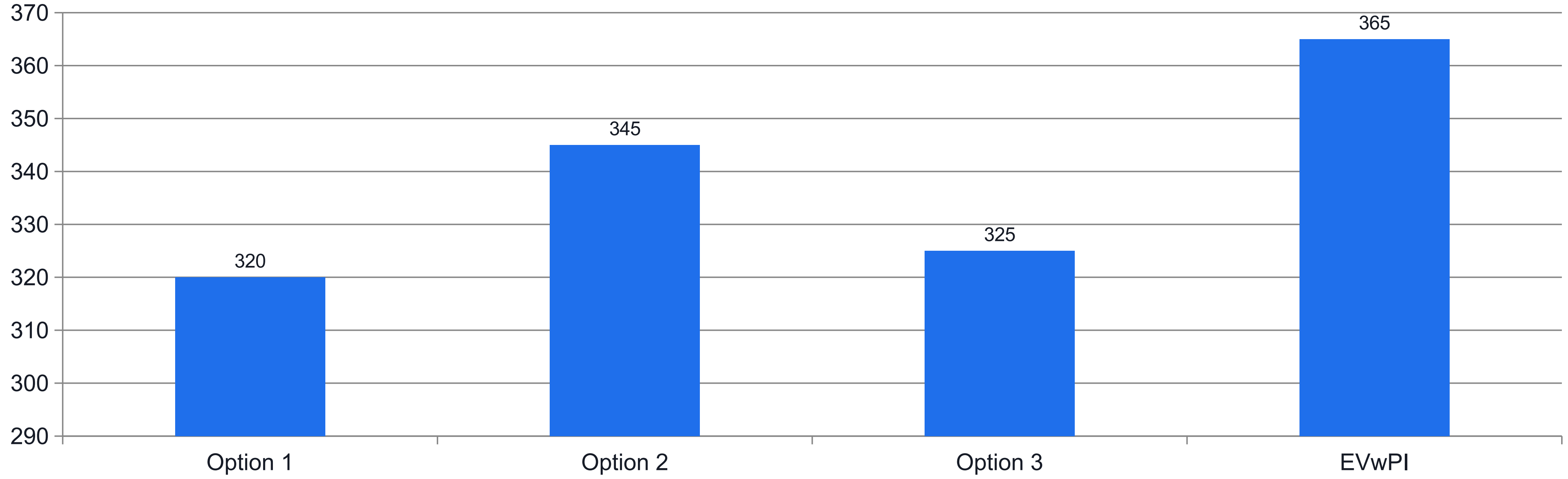
Lab 4 Evidence — EV, EVwPI and EVPI

Option	Fav (0.3)	Neu (0.5)	Unf (0.2)	EV ($'000)
Option 1 — Large	400	360	100	320
Option 2 — Medium	350	380	250	345
Option 3 — Small	300	390	200	325
Best in each state	400	390	250	EVwPI 365

Best EV (choose)   Best in state   EVwPI

EVIDENCE EV: 320 / 345 / 325 → choose Option 2. EVwPI = 0.3·400 + 0.5·390 + 0.2·250 = 365. EVPI = 365 − 345 = $20k.

Lab 4 Evidence — EV by Option vs EVwPI



WHAT THE DATA SHOWS

Option 2 has the highest EV ($345k). The gap up to EVwPI ($365k) is the $20k value of perfect information.

Expected Value, EVwPI and EVPI

✅ Expected result

The EV column shows Option 1 = 320, Option 2 = 345, Option 3 = 325; the best-EV flag points to Option 2; EVwPI reads 365 and EVPI reads 20 (all in $'000).

IF IT DOESN'T WORK — DIAGNOSE

Probabilities do not sum to 1

F4 must equal 1. A typo in C4:E4 skews every EV. Fix the probabilities first.

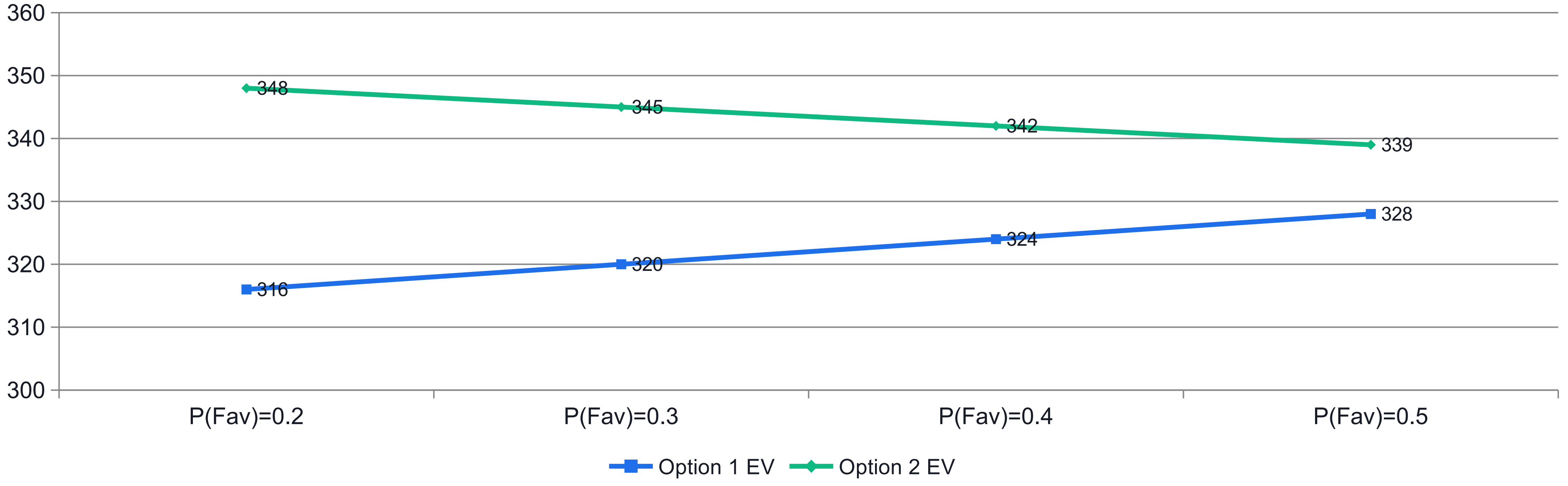
EV uses the wrong probability row

Lock the probability range with $C$4:$E$4 so SUMPRODUCT always multiplies by the correct state probabilities.

EVPI comes out negative

You subtracted in the wrong order or picked the wrong best EV. EVPI = EVwPI − best EV and is always ≥ 0.

How Sensitive Is the Choice to the Probabilities?



WHAT THE DATA SHOWS

Option 2 stays the EV-best choice across this range; only if a favourable market becomes very likely (P(Fav) > ~0.66) would Option 1 overtake — the recommendation is robust.

EVPI in Practice — Is a Market Study Worth It?

EVPI caps what perfect information is worth. Compare it to the price of a real, imperfect study.

```
# from Lab 4
EVPI = EVwPI - best EV
     = 365 - 345 = $20k

# a market study costs
     = $15k

# perfect info worth $20k
# an imperfect study < $20k
-> a $15k study can pay off
# never pay more than $20k
```

EVPI is the ceiling

$20k is the most any information is worth here.

Real studies are imperfect

Their value sits below EVPI.

Compare to cost

A $15k study can pay off; a $25k one cannot.

Decision rule

Never pay more than EVPI for information.

Why Expected Value Is Not the Whole Story

1. **EV ignores spread**
   Two options with the same EV can carry very different risk.

2. **Ruinous downside**
   A high-EV gamble with a small chance of ruin may be unacceptable.

3. **Attitude matters**
   Most managers are risk-averse — they trade some EV for certainty.

4. **Enter utility**
   Expected Utility replaces money with satisfaction to capture risk attitude.

Expected Utility — Capturing Risk Attitude

1

Utility function U(x)

Maps money to satisfaction — usually with diminishing returns.

2

Risk-averse (concave)

$U(x)=\sqrt{x}$: each extra dollar adds less utility — prefers certainty.

3

Expected utility

$EU = \Sigma\, p \cdot U(\text{payoff})$ — rank options by EU, not EV.

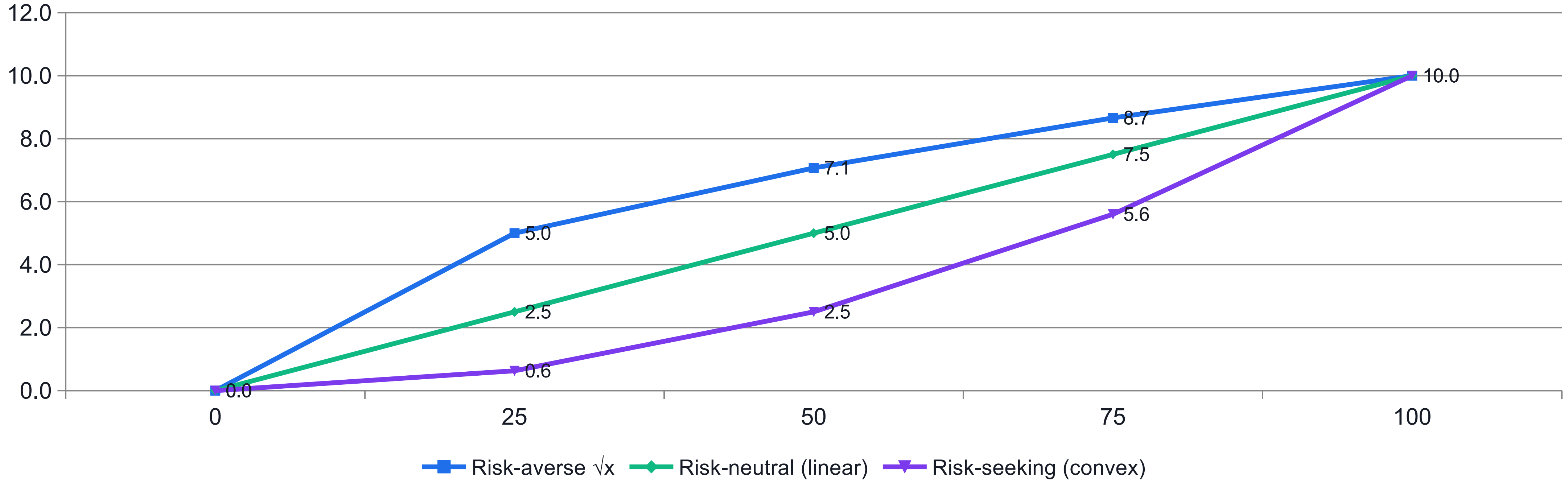
4

Certainty equivalent

$CE = U^{-1}(EU)$: the guaranteed sum worth the same as the gamble.

SHAPE = ATTITUDE

Utility Curves — Three Risk Attitudes



WHAT THE DATA SHOWS

A concave curve (√x) is risk-averse: it values a sure $50 above a 50/50 gamble on $0 or $100. Convex is risk-seeking; a straight line is risk-neutral (pure EV).

Utility, Certainty Equivalent & Risk Premium

EXPECTED UTILITY

$$EU = \Sigma\, p_i \cdot U(x_i)$$

average satisfaction

CERTAINTY EQUIVALENT

$$CE = U^{-1}(EU)$$

sure sum of equal worth

RISK PREMIUM

$$\text{Risk premium} = EV - CE$$

what risk costs you

READ IT AS for a risk-averse manager CE ≤ EV always; the gap is the risk premium.

Choosing a Utility Function

1. **Match the attitude**
   Concave = risk-averse, linear = neutral, convex = risk-seeking.

2. **Common choices**
   sqrt(x) and log(x) for risk aversion; 1 - e^(-x/R) with risk tolerance R.

3. **Calibrate it**
   Ask the manager for the certainty equivalent of one 50-50 gamble to fix the curve.

4. **Stay consistent**
   Use the same utility function across every option you compare.

Risk and Expected Utility

LAB 5

A6, K2 — review sufficiency and optimal utilisation when outcomes carry risk

Compare two resource-allocation projects with the SAME expected value but different risk, using a utility function to separate the risk-averse choice from a risk-neutral one, and quantify the risk premium.

YOU'LL BUILD

A utility table showing that with $U(x)=\sqrt{x}$ the safer Project A (certainty equivalent \$49.5k) beats the risky Project B (\$40.0k) even though both have EV=\$50k.

TOOLCHAIN

Microsoft Excel

DONE WHEN

$EV_A = EV_B = 50$, but $EU_A = 7.035 > EU_B = 6.325$ and $CE_A = 49.5 > CE_B = 40.0$, so the risk-averse choice is Project A; Project B carries a \$10k risk premium versus A's \$0.5k.

Risk and Expected Utility

1. **Set up two projects**
   Same EV=$50k: Project A pays…

2. **Choose a utility function**
   Risk-averse U(x)=√x maps…

3. **Compute expected utility**
   EU = Σ probability ×…

4. **Find certainty equivalents**
   CE = U^{-1}(EU): the guaranteed…

5. **Read off risk**
   Risk premium = EV − CE; the…

THE POINT

A6, K2 — review sufficiency and optimal utilisation when outcomes carry risk

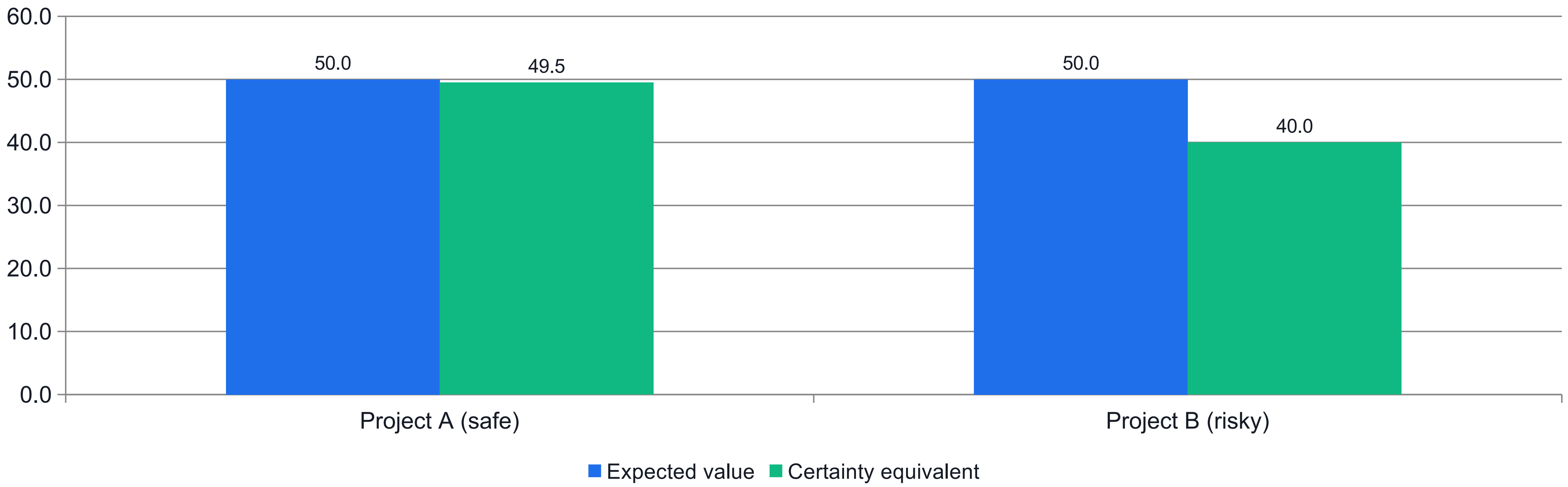
Lab 5 Evidence — Equal EV, Different Risk

Project	Outcomes ($'000)	EV	Exp. utility √	Certainty equiv.	Risk premium
A — Safe	40 / 60 (50-50)	50	7.035	49.5	0.5
B — Risky	10 / 90 (50-50)	50	6.325	40.0	10.0

Higher certainty equivalent → choose A

EVIDENCE Both projects have EV $50k, but A's certainty equivalent ($49.5k) beats B's ($40.0k). A risk-averse manager chooses A; B carries a $10k risk premium.

Lab 5 Evidence — EV Ties, Certainty Equivalent Decides



WHAT THE DATA SHOWS

EV cannot tell the projects apart; the certainty equivalent can — and it points clearly to the safer Project A.

Risk and Expected Utility

✅ Expected result

$EV_A = EV_B = 50$, but $EU_A = 7.035 > EU_B = 6.325$ and $CE_A = 49.5 > CE_B = 40.0$, so the risk-averse choice is Project A; Project B carries a $10k risk premium versus A's $0.5k.

IF IT DOESN'T WORK — DIAGNOSE

Both projects tie

You compared EV, not expected utility. With equal EV the tie-break is EU / certainty equivalent, not EV.

Utility applied to the EV instead of each outcome

Apply U(x) to every outcome first, then average — $U(E[x]) \neq E[U(x)]$ (that gap is exactly the risk premium).

Certainty equivalent exceeds EV

For a risk-averse (concave) utility, CE ≤ EV always. A CE above EV means the inverse was applied wrongly.

Recap — Decision Analysis

- A decision problem = decisions × uncertain states (with probabilities) → payoffs; a decision tree makes it visible.
- Expected Value ranks the options: KAZO's best is Option 2 at $345k.
- EVwPI ($365k) values perfect foresight; EVPI ($20k) is the most to pay for perfect information.
- Expected Value ignores risk; Expected Utility and the certainty equivalent capture a risk-averse attitude.
- Equal-EV projects can differ sharply once risk is priced — Project A (CE $49.5k) beats Project B (CE $40.0k).

03

TOPIC 03

Multi-Criteria Decision Analysis

Weighted scoring board · Analytic Hierarchy Process · pairwise comparison, consistency and KPIs

Key Concepts — Multi-Criteria Decision Analysis

1

Real decisions weigh many criteria

When cost, quality, risk and fit all matter, a single objective is not enough — you need a multi-criteria method.

2

The scoring board is the quick tool

Weight each criterion, rate each option, and rank by the weighted total score = Σ weight × rating.

3

AHP adds rigour through pairwise comparison

Compare criteria two at a time on Saaty's 1-9 scale and derive priority weights from the comparison matrix.

4

Consistency is checkable

The Consistency Ratio CR = CI/RI tests whether your judgements hang together; CR ≤ 0.10 is acceptable.

5

AHP synthesises a defensible ranking and KPIs

Combine criteria priorities with per-criterion scores to rank options, and reuse the weights as resource-management KPIs.

THE MULTI-CRITERIA PROBLEM

When One Number Isn't Enough

1

Many criteria

Real choices weigh cost, quality, delivery, risk and fit at once.

2

Different units

Dollars, days, ratings — not directly comparable.

3

Trade-offs

The cheapest option is rarely the best on quality.

4

MCDA

Multi-Criteria Decision Analysis combines criteria into one defensible ranking.

FIVE STEPS

The MCDA Process

1
List criteria
What matters for this decision

→

2
Weight them
How important each criterion…

→

3
Score options
Rate every option per…

→

4
Combine
Weighted total per option

→

5
Rank & review
Recommend and stress-test

THE POINT

A transparent, repeatable path from many criteria to one ranking you can defend.

THE QUICK MCDA TOOL

The Weighted Scoring Board

SCORE

$$\text{Score(option)} = \Sigma\ w_j \cdot r_j$$

weight × rating, summed

WEIGHTS

w_j = importance of criterion j

set by the decision-maker

RATINGS

r_j = how the option scores on j

a common 1-5 scale

READ IT AS rank options by weighted total; keep every criterion on the same scale.

HOW IT'S BUILT

The Scoring Board Layout

Criterion	Weight	Option A	Option B
Criterion 1	w1	r · w	r · w
Criterion 2	w2	r · w	r · w
Weighted total		Σ w·r	Σ w·r

Weighted total = SUMPRODUCT

EVIDENCE Each option's total is =SUMPRODUCT(weights, ratings). The highest total wins.

TOPIC 03 · HANDS-ON LAB 6

Weighted Scoring Board (MCDA)

LAB 6

A7, K3 — propose improvements by ranking options against several weighted criteria

Rank three suppliers against five weighted criteria using a scoring board: weight each criterion, rate each supplier, and compute the weighted total score.

YOU'LL BUILD

A scoring board that ranks Supplier A (42) above B (41) and C (40), with a sensitivity check on the criterion weights.

TOOLCHAIN

Microsoft Excel

DONE WHEN

The weighted totals read Supplier A = 42, Supplier B = 41, Supplier C = 40, so Supplier A ranks first; raising the Price weight shifts the ranking toward Supplier B.

Weighted Scoring Board (MCDA)

1. **List criteria & weights**
   Price, Quality, Lead time…

2. **Rate each option**
   Score every supplier 1-5 on…

3. **Weight the ratings**
   Weighted cell = weight ×…

4. **Total and rank**
   Total score = Σ weight ×…

5. **Sensitivity check**
   Change a weight and see…

THE POINT

A7, K3 — propose improvements by ranking options against several weighted criteria

LAB 6 · WEIGHTED SCORING

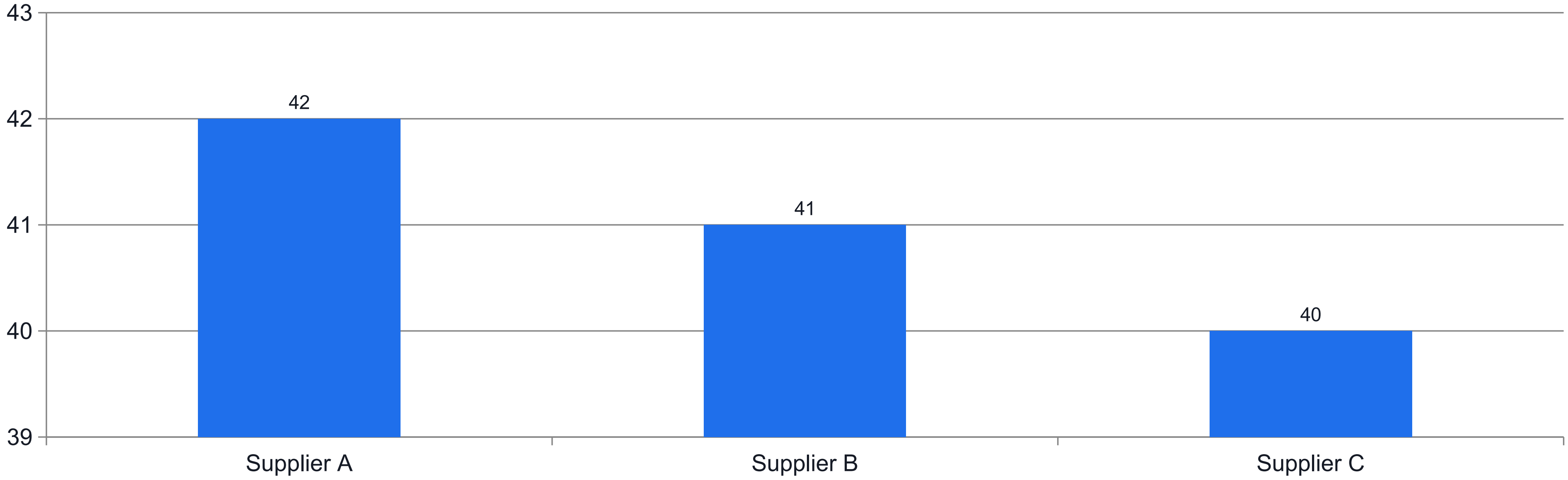
Lab 6 Evidence — Supplier Scoring Board

Criterion	Weight	Supplier A	Supplier B	Supplier C
Price	3	3	5	4
Quality	3	5	3	4
Lead time	2	3	4	2
Reliability	2	4	3	5
Support	1	4	3	2
Weighted total		42	41	40

Winner (42)   Weighted totals

EVIDENCE Supplier A wins (42) — but by a single point over B. Close totals are exactly why AHP's pairwise weighting is used next.

Lab 6 Evidence — Supplier Totals & Weight Sensitivity



WHAT THE DATA SHOWS

With only one point separating the top two, raising the Price weight can flip the ranking to Supplier B — the result is sensitive to the weights.

Weighted Scoring Board (MCDA)

✅ Expected result

The weighted totals read Supplier A = 42, Supplier B = 41, Supplier C = 40, so Supplier A ranks first; raising the Price weight shifts the ranking toward Supplier B.

IF IT DOESN'T WORK — DIAGNOSE

All suppliers score the same

The weights are all equal or all 1. Differentiate the weights to reflect real priorities.

Ratings on different scales

Keep every criterion on the same 1-5 scale, or normalise first — mixing scales silently biases the total.

Ranking feels arbitrary

Scores are close (42/41/40). That fragility is exactly why AHP's pairwise weighting is used next.

Normalising Weights & Ratings

1

Weights sum to 1

Divide each weight by the total so priorities are comparable.

2

One rating scale

Keep every criterion on the same scale (e.g. 1-5) before weighting.

3

Reverse cost criteria

For 'lower is better', map cheapest to the highest score.

4

Why it matters

Un-normalised scores let a large-numbered criterion dominate by accident.

Scoring Board — Weight Sensitivity

Price weight	Supplier A	Supplier B	Winner
3 (base)	42	41	Supplier A
4	45	46	Supplier B
5	48	51	Supplier B

WHY IT MATTERS raising the Price weight, where B leads, flips the winner — a reason to derive weights rigorously with AHP.

WHY AHP

From Scoring Boards to AHP

1

Weights feel arbitrary

Where did 'Price = 3' come from? Scoring boards rarely justify the weights.

2

Pairwise is easier

People compare two things at a time more reliably than rating all at once.

3

AHP derives weights

The Analytic Hierarchy Process turns pairwise judgements into priority weights.

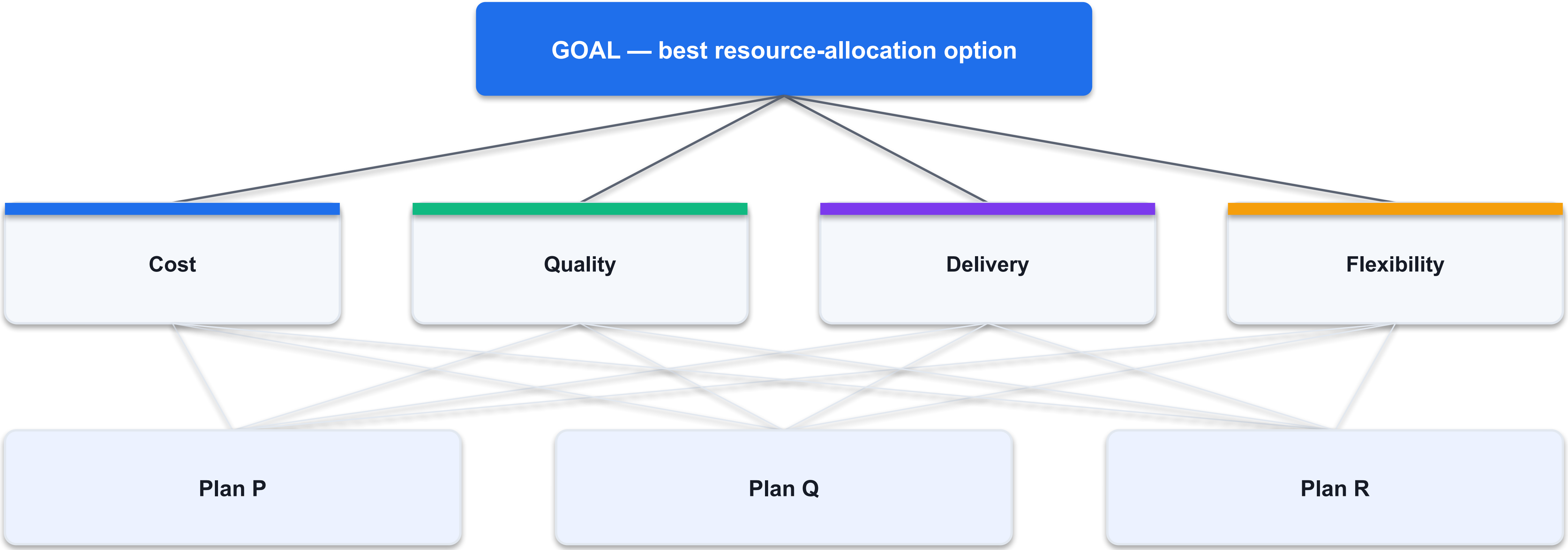
4

And checks them

AHP measures whether your judgements are consistent (the CR).

GOAL → CRITERIA → ALTERNATIVES

The AHP Hierarchy



AHP structures the decision top-down: weight the criteria, score the alternatives per criterion, then synthesise.

THE AHP WORKFLOW

The Three Parts of AHP

1

Prioritise criteria

Pairwise-compare the criteria…

→

2

Check consistency

Consistency Ratio CR ≤ 0.10

→

3

Synthesise

Combine to rank the…

THE POINT

AHP produces weights AND a consistency check AND a final ranking — all traceable.

HOW TO COMPARE TWO CRITERIA

Saaty's 1-9 Pairwise Scale

Judgement	Value	Meaning
Equal	1	Both criteria matter equally
Moderate	3	One is moderately more important
Strong	5	One is strongly more important
Very strong	7	One is very strongly more important
Extreme	9	One utterly dominates
Reciprocal	1/n	The reverse comparison

WHY IT MATTERS if Cost is 'moderately' more important than Quality, enter 3 — and 1/3 in the mirror cell.

STEP 1 · JUDGEMENTS ON THE 1-9 SCALE

The Criteria Pairwise Matrix

Criterion	Cost	Quality	Delivery	Flexibility
Cost	1	3	2	2
Quality	1/3	1	1/4	1/4
Delivery	1/2	4	1	1/2
Flexibility	1/2	4	2	1

HOW TO READ read the row against the column: Cost vs Quality = 3 (moderately more important); the mirror cell is 1/3.

STEP 2 · FROM MATRIX TO WEIGHTS

Deriving the Priority Weights

Normalise each column (divide by its total), then average each row — that row average is the criterion's priority weight.

```
# 1. column totals
Cost 2.333  Quality 12
Delivery 5.25  Flex 3.75

# 2. normalise each cell
cell / column total

# 3. priority = row average
Cost       = 0.398
Quality    = 0.085
Delivery   = 0.218
Flex       = 0.299
sum        = 1.000
```

Column totals

Add each column of the pairwise matrix.

Normalise

Divide every cell by its column total so each column sums to 1.

Row average

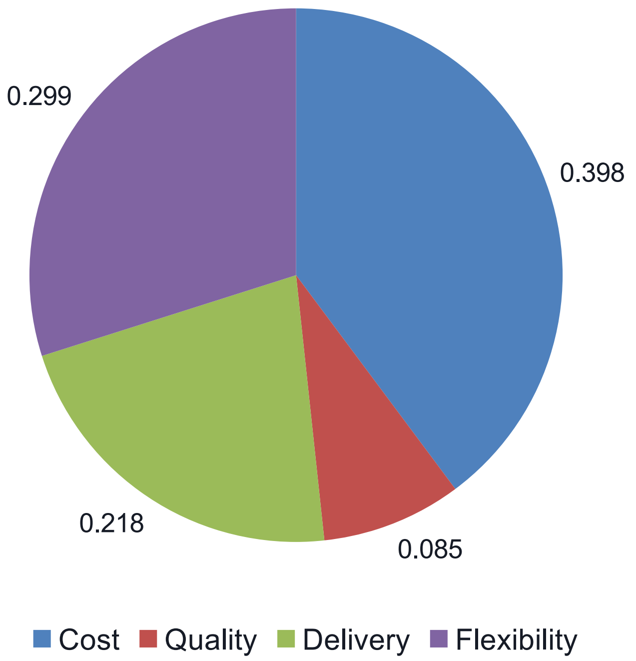
Average each normalised row → the priority weight.

Sanity check

The four priorities sum to 1.000.

THE DERIVED WEIGHTS

Criteria Priorities



WHAT THE DATA SHOWS

Cost dominates at 0.398 (about 40%); Quality is least important at 0.085. These weights drive the final ranking.

AHP — Criteria Priorities and Consistency

LAB 7

A8, K3 — refine the plan by deriving defensible criteria weights and checking their consistency

Use the Analytic Hierarchy Process to weight four resource-management criteria by pairwise comparison on Saaty's 1-9 scale, derive priority weights, and verify judgement consistency with the Consistency Ratio.

YOU'LL BUILD

Criteria priorities Cost 0.398, Quality 0.085, Delivery 0.218, Flexibility 0.299, with lambda_max=4.185, CI=0.062 and CR=0.068 (consistent).

TOOLCHAIN

Microsoft Excel

DONE WHEN

The priority vector reads Cost 0.398, Quality 0.085, Delivery 0.218, Flexibility 0.299 (sum 1.000); lambda_max = 4.185, CI = 0.062, CR = 0.068 which is ≤ 0.10, so the criteria weights are consistent and usable.

AHP — Criteria Priorities and Consistency

1. **Build the pairwise matrix**
   Compare each pair of criteria…
2. **Normalise the columns**
   Divide each cell by its…
3. **Derive priorities**
   Priority of a criterion = the…
4. **Compute lambda_max**
   Weighted-sum vector ÷…
5. **Check consistency**
   CR = CI/RI must be ≤ 0.10 to…

THE POINT

A8, K3 — refine the plan by deriving defensible criteria weights and checking their consistency

Lab 7 Evidence — Criteria Matrix & Priorities

Criterion	Cost	Quality	Delivery	Flexibility	Priority
Cost	1	3	2	2	**0.398**
Quality	1/3	1	1/4	1/4	**0.085**
Delivery	1/2	4	1	1/2	**0.218**
Flexibility	1/2	4	2	1	**0.299**

HOW TO READ the priority column is the row average of the column-normalised matrix; the four weights sum to 1.000.

Lab 7 Evidence — Checking Consistency

ΛMAX

$$\lambda_{max} = \text{avg}\left((A\cdot w)_i / w_i \right) = 4.185$$

≥ n when consistent

CONSISTENCY INDEX

$$CI = (\lambda_{max} - n)/(n - 1) = 0.062$$

n = 4

CONSISTENCY RATIO

$$CR = CI / RI = 0.062 / 0.90 = 0.068$$

RI(4) = 0.90

READ IT AS $CR = 0.068 \le 0.10$ — the judgements are consistent, so the weights are trustworthy.

The Random Index (RI) Table

n (criteria)	3	4	5	6	7	8
Random Index RI	0.58	0.90	1.12	1.24	1.32	1.41

WHY IT MATTERS RI is the average CI of random reciprocal matrices of size n. For 4 criteria, RI = 0.90.

Consistency Gate

Is the Consistency Ratio CR ≤ 0.10 ?

Yes — accept the weights

Judgements hang together. Use the priorities to score and rank the alternatives.

No — revise judgements

Find the most inconsistent pair, re-judge it on the 1-9 scale, and recompute CR.

CR ≤ 0.10 keeps AHP weights trustworthy without demanding perfect (CR = 0) consistency.

LAB 7 · VERIFY & DIAGNOSE

AHP — Criteria Priorities and Consistency

✅ Expected result

The priority vector reads Cost 0.398, Quality 0.085, Delivery 0.218, Flexibility 0.299 (sum 1.000); lambda_max = 4.185, CI = 0.062, CR = 0.068 which is ≤ 0.10, so the criteria weights are consistent and usable.

IF IT DOESN'T WORK — DIAGNOSE

Priorities do not sum to 1

You averaged the raw matrix, not the column-normalised matrix. Normalise each column first, then average the rows.

CR is above 0.10

Judgements conflict (e.g. A»B, B»C but A≈C). Revisit the largest inconsistency and re-judge that pair.

Matrix is not reciprocal

Every lower-triangle cell must be 1/(its mirror). Enter reciprocals with =1/cell, never by hand.

AHP vs the Scoring Board

Aspect	Scoring board	AHP
Weights	Set directly (arbitrary)	Derived from pairwise comparison
Consistency	Not checked	Consistency Ratio <= 0.10
Effort	Low	Higher (more comparisons)
Defensibility	Moderate	High — every weight is justified

WHY IT MATTERS use a scoring board for a quick call; use AHP when the decision must be defended or the criteria conflict.

Alternative-Level Pairwise (Criterion: Cost)

Criterion	Plan P	Plan Q	Plan R	Priority
Plan P	1	2	3	0.54
Plan Q	1/2	1	2	0.297
Plan R	1/3	1/2	1	0.163

HOW TO READ AHP repeats the pairwise process at the alternative level, once per criterion, to get each option's local priority.

THE FINAL AHP STEP

Synthesis — Combining Criteria & Alternatives

GLOBAL SCORE	$\text{Global(option)} = \Sigma_j \; W_j \cdot s_j\text{(option)}$	criteria weight × local score
LOCAL SCORE	s_j = the option's normalised score on criterion j	column sums to 1
RANK	Highest global score = recommended allocation	fully traceable

READ IT AS each option's global priority is the weighted sum of its per-criterion scores.

AHP Synthesis — The Arithmetic

Multiply each criterion weight by the plan's local score on that criterion, then sum.

```
# criteria weights (Lab 7)
Cost .398  Quality .085
Delivery .218  Flex .299

# Plan P local scores
.54  .16  .30  .33

# global = sum(w * local)
.398*.54 + .085*.16
  + .218*.30 + .299*.33
= 0.393   (rank 1)
```

Per criterion

weight x the plan's local score.

Sum across criteria

gives the plan's global priority.

Plan P = 0.393

the highest of the three plans.

Traceable

every number traces back to a judgement.

Integrated Resource-Allocation Evaluation & KPIs

LAB 8

A8, A9, K3 — synthesise a defensible ranking and convert the criteria weights into resource-management KPIs

Combine the AHP criteria priorities with per-criterion scores for three resource-allocation plans to produce a final synthesised ranking, then turn the criteria weights into monitoring KPIs.

YOU'LL BUILD

A synthesis table ranking Plan P (0.393) above Plan R (0.309) and Plan Q (0.299), plus a KPI dashboard weighting Cost 40%, Flexibility 30%, Delivery 22%, Quality 8%.

TOOLCHAIN

Microsoft Excel

DONE WHEN

The synthesis table ranks Plan P = 0.393 first, then Plan R = 0.309 and Plan Q = 0.299 (sum ≈ 1.000); the KPI dashboard carries the AHP-derived weights (Cost 40%, Flexibility 30%, Delivery 22%, Quality 8%) each mapped to a measurable KPI.

Integrated Resource-Allocation Evaluation & KPIs

1 — Bring in criteria weights

Reuse the AHP priorities…

2 — Score plans per criterion

Normalised local scores for…

3 — Synthesise

Global score = Σ criteria…

4 — Rank the plans

Highest global score is the…

5 — Set KPIs

Reuse the criteria weights as…

THE POINT

A8, A9, K3 — synthesise a defensible ranking and convert the criteria weights into resource-management KPIs

Lab 8 Evidence — Integrated Synthesis

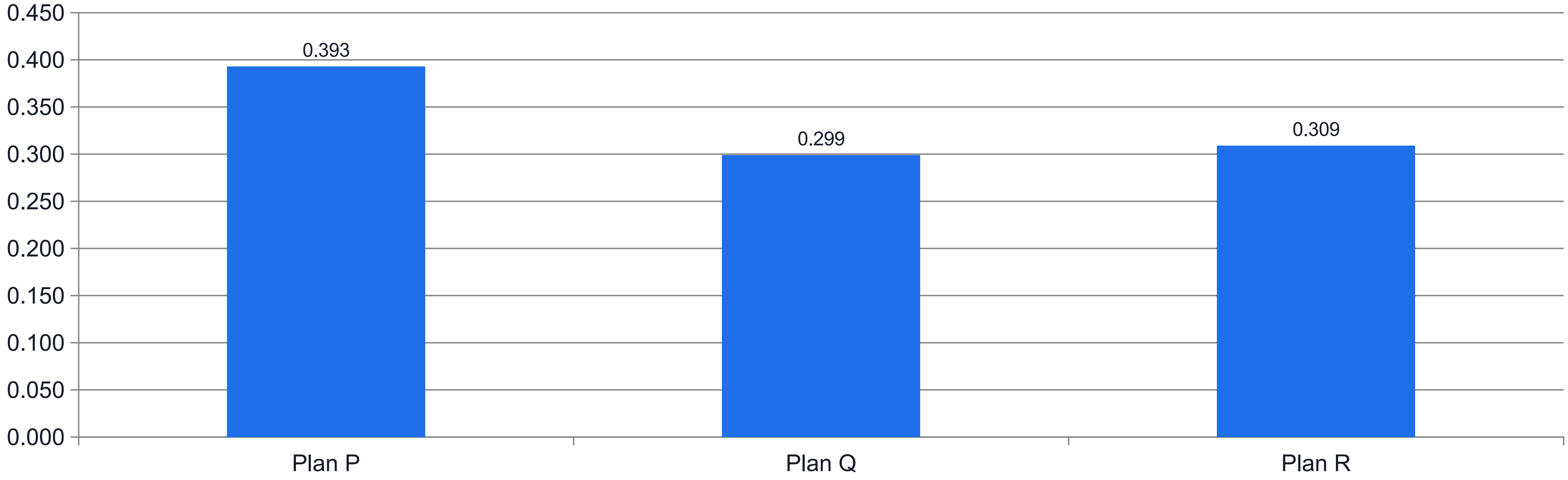
Criterion	Weight	Plan P	Plan Q	Plan R
Cost	0.398	0.54	0.30	0.16
Quality	0.085	0.16	0.54	0.30
Delivery	0.218	0.30	0.16	0.54
Flexibility	0.299	0.33	0.33	0.34
Global score		0.393	0.299	0.309

Winner — Plan P

Global priorities

EVIDENCE Global = Σ weight × local score. Plan P wins (0.393); the ranking is fully traceable to the AHP criteria weights.

Lab 8 Evidence — Final Allocation Ranking



WHAT THE DATA SHOWS

Plan P is the recommended resource allocation, ahead of Plan R and Plan Q — a defensible, weight-driven decision.

LAB 8 · A9 — KPIs FROM THE SAME WEIGHTS

Resource-Management KPIs from the AHP Weights

1. **Cost — 40%**
   Cost variance vs budget; target ≤ 5% over.

2. **Flexibility — 30%**
   Time to re-allocate resources; target < 2 days.

3. **Delivery — 22%**
   On-time delivery rate; target ≥ 95%.

4. **Quality — 8%**
   Defect / rework rate; target ≤ 2%.

The Monitor-Review-Reallocate Loop

1

Set KPIs

Weighted by the AHP priorities

→

2

Monitor

Measure each KPI on a cadence

→

3

Review

Compare to targets &…

→

4

Reallocate

Shift resources when a…

THE POINT

Choosing the plan is not the end — KPIs monitor usage and trigger re-allocation, closing the resource-management loop.

LAB 8 · VERIFY & DIAGNOSE

Integrated Resource-Allocation Evaluation & KPIs

✅ Expected result

The synthesis table ranks Plan P = 0.393 first, then Plan R = 0.309 and Plan Q = 0.299 (sum ≈ 1.000); the KPI dashboard carries the AHP-derived weights (Cost 40%, Flexibility 30%, Delivery 22%, Quality 8%) each mapped to a measurable KPI.

IF IT DOESN'T WORK — DIAGNOSE

Global scores do not sum to ~1

Either the criteria weights or a local-score column does not sum to 1. Re-normalise before synthesising.

A plan wins on one criterion but loses overall

That is correct behaviour — synthesis balances all criteria by weight, not a single strength.

KPI weights ignore AHP

Derive KPI weights from the AHP priorities so monitoring reflects the same priorities used to choose the plan.

Choosing the Right Method

Is the goal a single quantity to optimise under hard limits?

Yes → Linear Programming

One objective, linear resource constraints, known data — formulate and run Solver.

No → Decision Analysis or MCDA

Uncertainty with probabilities → Decision Analysis. Several weighted criteria → MCDA / AHP.

the three methods are complementary tools — match the method to the shape of the decision.

ONE TOOLKIT, THREE TOOLS

LP vs Decision Analysis vs MCDA

Method	Best when	Key output	Course tool
Linear Programming	One objective + hard resource limits	Optimal allocation + shadow prices	Excel Solver
Decision Analysis	Uncertain future with probabilities	Best EV; value of information	Decision tree / EV
MCDA / AHP	Several weighted, conflicting criteria	Ranked options + KPI weights	Scoring board / AHP

WHY IT MATTERS many real projects chain them: LP to size the plan, decision analysis to handle uncertainty, AHP to weigh qualitative criteria.

TOPIC 03 · AVOID THESE

Common AHP Pitfalls

⚠ Non-reciprocal matrix

Fix: Enter mirror cells as =1/cell, never by hand.

⚠ Ignoring a high CR

Fix: CR > 0.10 means re-judge — do not ship inconsistent weights.

⚠ Too many criteria

Fix: Keep to about seven; more makes pairwise comparison unreliable.

⚠ Averaging raw scores

Fix: Normalise columns before averaging rows, or the priorities are wrong.

Group AHP — Aggregating Judgements

1. **Whose weights?**
Real decisions involve several stakeholders with different priorities.

2. **Aggregate judgements**
Combine individual pairwise judgements with the geometric mean.

3. **Or aggregate priorities**
Average each person's derived priority vector.

4. **Check consistency**
Test the group matrix's CR just as for one person.

Consistency Repair — Fixing a Bad Judgement

If CR > 0.10, find the pair that conflicts most with the derived priorities and re-judge it.

```
# suppose CR = 0.18  (>0.10)
# priorities imply
Cost/Delivery ~ .398/.218
            ~ 1.8
# but you entered
Cost vs Delivery = 5  (too high)

# revise toward 2
Cost vs Delivery = 2
-> recompute -> CR = .06  (ok)
```

Spot the outlier

The judgement most at odds with the priority ratios.

Compare to implied

Priorities imply Cost ~1.8x Delivery, not 5x.

Re-judge that pair

Move it toward the implied ratio.

Recompute CR

One fix usually brings CR under 0.10.

Recap — Multi-Criteria Decision Analysis

- MCDA combines several weighted criteria into one defensible ranking.
- The scoring board (Score = $\Sigma\, w \cdot r$) is quick; Supplier A won at 42 — but only just.
- AHP derives criteria weights from pairwise comparison: Cost 0.398, Quality 0.085, Delivery 0.218, Flexibility 0.299.
- The Consistency Ratio ($CR = 0.068 \le 0.10$) confirms the judgements hang together.
- Synthesis ranks Plan P first (0.393); the same weights become resource-management KPIs (A9).

WRAP-UP

Course Summary & Next Steps

LEARNING OUTCOMES

What You Achieved

1

LO1 · Linear Programming

Formulated LP models and used Excel Solver to maximise profit, then read binding constraints, shadow prices and sensitivity ranges.

2

LO2 · Decision Analysis

Built decision trees, ranked options by Expected Value, valued information (EVPI) and priced risk with Expected Utility.

3

LO3 · Multi-Criteria Decision Analysis

Ran a scoring board, derived AHP weights with a consistency check, synthesised a ranking and set KPIs.

NEXT STEPS

Continue Your Learning

1

Rebuild every lab

Redo each workbook from a blank sheet until Solver, EV and AHP are automatic.

2

Apply it at work

Model one real allocation, decision or supplier choice from your own organisation.

3

Go deeper

Integer programming, Monte-Carlo risk, and AHP with more criteria and alternatives.

4

Combine methods

Chain LP → decision analysis → AHP on a single project for an end-to-end plan.

5

Keep the toolkit

The Excel Solver, decision trees and AHP templates work on any resource problem.

6

Ask us

Bring your own data to a follow-up clinic or the community forum.

APPLY IT AT WORK

Your 30-Day Action Plan

1

Week 1

Rebuild Labs 1-3: model one product-mix problem from your own work in Solver.

2

Week 2

Rebuild Labs 4-5: draw a decision tree for a real uncertain choice and price the risk.

3

Week 3

Rebuild Labs 6-8: run an AHP on a supplier or project selection you face.

4

Week 4

Set KPIs from your AHP weights and start the monitor-review loop.

Practice Exam

exams.tertiaryinfotech.com

Sharpen your readiness before the WA + PP

1. **Linear Programming**
   Formulate models, run Solver, read binding constraints & shadow prices.

2. **Decision Analysis**
   Expected Value, EVwPI / EVPI and Expected-Utility questions.

3. **Multi-Criteria Decision Analysis**
   Scoring boards, AHP priorities and the Consistency Ratio.

4. **Timed mock**
   Attempt under exam conditions, then review every explanation.

CONTINUE WITH TERTIARY INFOTECH

Recommended Courses

1. WSQ Data Analysis and Visualisation with Microsoft Excel
2. WSQ Business Statistics and Data Analytics with Excel
3. WSQ Data Analytics and Visualisation with Power BI
4. WSQ Project Management Essentials
5. WSQ Python for Data Analysis and Automation

One Toolkit, Three Decisions

1. **Optimise -> Linear Programming**
   Maximise profit or minimise cost within hard resource limits.

2. **Decide under risk -> Decision Analysis**
   Rank options by expected value and price uncertainty.

3. **Evaluate -> MCDA / AHP**
   Weigh conflicting criteria into a defensible ranking and KPIs.

WE'RE HERE TO HELP

Support

- If you have any enquiries during or after the class, contact us below.
- Email: enquiry@tertiaryinfotech.com
- Tel: +65 6100 0613
- Website: www.tertiarycourses.com.sg
- LMS / TMS: https://lms-tms.tertiaryinfotech.com

Assessment

1

Written Assessment (SAQ)

1 hour · open book · short-answer knowledge questions (K1-K3).

2

Practical Performance (PP)

1 hour · open book · hands-on Excel tasks across LP, DA and MCDA (A1-A9).

3

Digital attendance

Take the Assessment digital attendance (TRAQOM) before you start.

4

Submit on the LMS

Save your workbooks and upload your answers at lms-tms.tertiaryinfotech.com.

ON ASSESSMENT DAY

Assessment Flow

1

TRAQOM survey

Scan the QR code on the LMS

→

2

Digital attendance

Scan the SSG assessment QR

→

3

Sit WA then PP

Open book · 1 hour each

→

4

Submit on the LMS

Upload your answers &…

→

5

Sign the record

Sign the Assessment Summary…

REMEMBER

All five steps are mandatory for WSQ funding — missing the digital attendance or the TRAQOM survey can invalidate your claim.

Digital Attendance (Mandatory)

1

Three times a day

Take the AM, PM and Assessment digital attendance — mandatory for every WSQ-funded course.

2

Trainer shows the QR

The trainer or administrator displays the SSG digital-attendance QR code (TRAQOM).

3

Scan and submit

Scan the QR with your phone camera and submit your attendance.

4

75% minimum

A minimum of 75% attendance is required to be eligible for assessment and funding.

WELL DONE

Thank You!

You can now optimise resources with Linear Programming, decide under uncertainty with Decision Analysis, and evaluate with Multi-Criteria Decision Analysis — all in Excel.